



TRINITY GRAMMAR SCHOOL

Mathematics Department

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(NESA Student Number | Year 12 only)

2022

Year 12

Mathematics Extension 1

TRIAL EXAMINATION

HSC ASSESSMENT TASK 4

Date of Assessment Task: Friday, 26 August 2022

General

Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Erasable pens and all electronic devices (or timekeepers), unless otherwise advised, are not permitted in this Assessment
- NESA approved calculators may be used
- A formula and data sheet is provided
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Write your NESA Student Number and your Class teacher on the question paper and on any answer sheets or writing booklets used to write your responses to the questions submitted
- If you do not attempt a question you must submit an answer sheet or writing booklet for that question clearly indicating N/A and your NESA Student Number or Name

Total marks:

70

Section I – 10 marks (pages 2 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 8 – 15)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

- **HSC Assessment Weighting:** 30%

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 Vectors **a**, **b** and **c** are shown below.

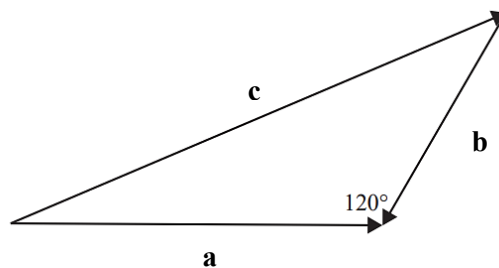
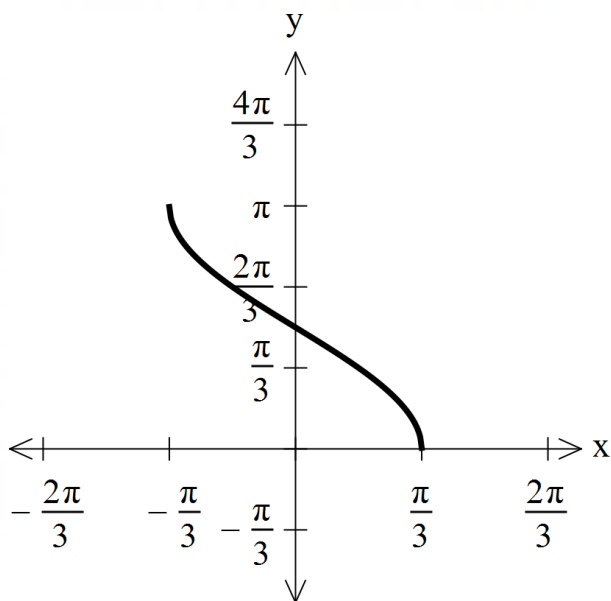


diagram not to scale

Which statement is correct?

- A. $|\mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2$
- B. $|\mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - |\mathbf{ab}|$
- C. $|\mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - |\mathbf{a}||\mathbf{b}|$
- D. $|\mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{a}||\mathbf{b}|$
- 2 What is the gradient of the tangent at $(7, 2)$ on the **inverse** function of $f(x) = x^3 - 1$?
- A. $\frac{1}{147}$
- B. $\frac{1}{12}$
- C. 12
- D. 147

- 3 The equation of the graph drawn below is in the form $y = A \cos^{-1}(Bx)$ where $A, B \in \mathbb{R}$. Which of the following represents the equation of the graph drawn below?



- A. $y = \cos^{-1}\left(\frac{3x}{\pi}\right)$
- B. $y = \cos^{-1}\left(\frac{\pi x}{3}\right)$
- C. $y = \frac{\pi}{3} \cos^{-1}(x)$
- D. $y = \frac{3}{\pi} \cos^{-1}(x)$

- 4 Consider the vectors $\mathbf{a} = x\mathbf{i} + \mathbf{j}$, $\mathbf{b} = \mathbf{i} - \mathbf{j}$, $\mathbf{c} = \mathbf{i} + x\mathbf{j}$ where x is a constant.
 θ is the angle between vectors $\mathbf{a}$ and $\mathbf{b}$, and φ is the angle between vectors $\mathbf{b}$ and $\mathbf{c}$.
What is the value of $\cos \theta \cos \varphi$?

A. $-\frac{(x+1)^2}{2(1+x^2)}$

B. $-\frac{(x-1)^2}{2(1+x^2)}$

C. $\frac{\sqrt{2}(1-x^2)}{(1+x^2)}$

D. $\frac{\sqrt{2}(1+x^2)}{(1-x^2)}$

- 5 A box contains 28 red balls, 17 yellow balls and 21 green balls.
What is the smallest number n (where n is a positive integer) such that any n balls drawn from the box will contain at least 15 balls of the same colour?

A. 43

B. 45

C. 46

D. 66

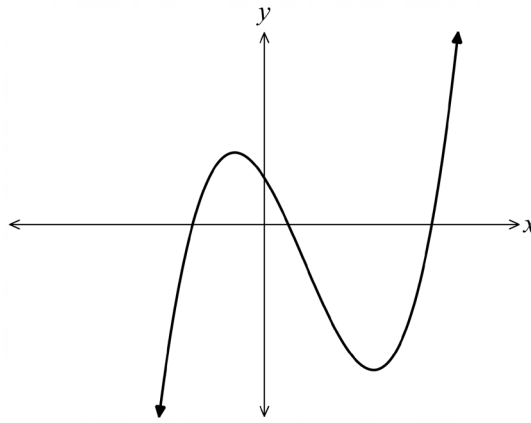
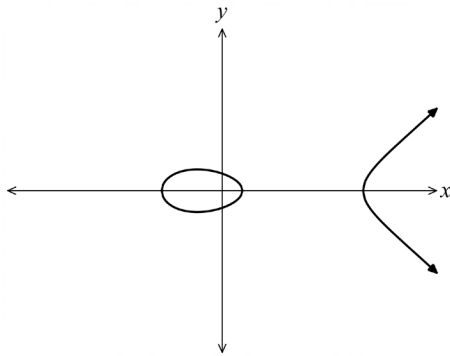


diagram not to scale

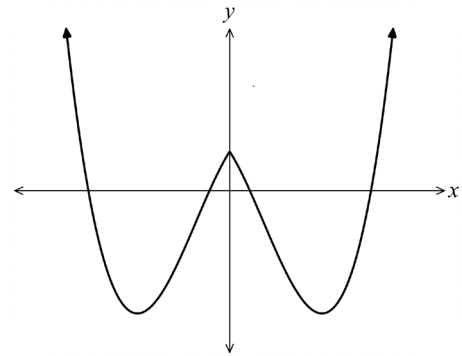
The graph of $y = f(x)$ is shown above.

Which of the following is a possible graph for $y^2 = f(x)$?

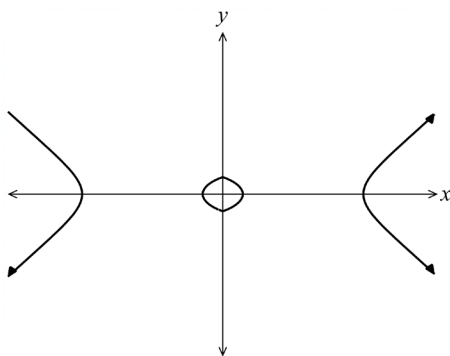
A.



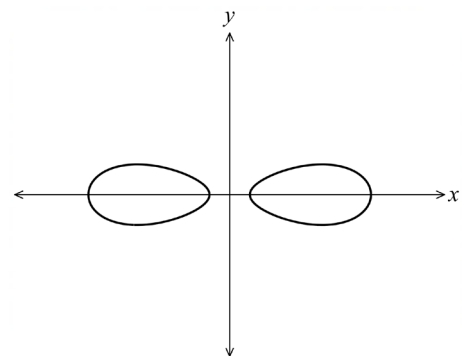
B.



C.



D.



- 7 When $x^4 + 2x^3 - 5x^2 + x + 3$ is divided by $x^2 + x - 2$, the remainder is in the form $Ax + B$. What is the sum of A and B ?

- A. -5
- B. -3
- C. 1
- D. 2

- 8 Which of the following represents the volume of the solid of revolution formed by rotating the graph of $y = \sqrt{9 - (x - 1)^2}$ about the x -axis?

- A. $4\pi(3)^2$
- B. $\pi \int_{-2}^4 \left(\sqrt{9 - (x - 1)^2} \right) dx$
- C. $\pi \int_{-2}^4 (9 - (x - 1)^2)^2 dx$
- D. $\pi \int_{-4}^2 (9 - (x - 1)^2) dx$

- 9 A curve is defined parametrically by $x = \sec(t) + 1$ and $y = \tan(t)$ where $t \in [0, \frac{\pi}{2})$. Which of the following statements is correct for the Cartesian equation **and** domain of the curve?

- A. $y = \sqrt{x^2 - 2x}$, $x \in [2, \infty)$
- B. $y = \sqrt{x^2 - 2x}$, $x \in (-\infty, 0] \cup [2, \infty)$
- C. $y^2 = x^2 - 2x$, $x \in [2, \infty)$
- D. $y^2 = x^2 - 2x$, $x \in (-\infty, 0] \cup [2, \infty)$

- 10 Sand falls from a chute to form a pile in the shape of a right circular cone of radius r , with a semi-vertex angle 60° as shown below. Sand is added to the pile at a rate of 1.5 m^3 per minute.

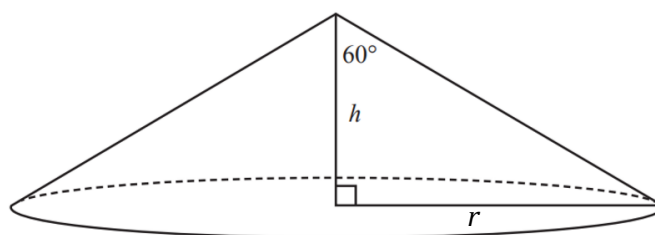


diagram not to scale

When the height of the pile, h , is 0.5 metres, what is the approximate instantaneous rate of change in height, in metres per minute?

- A. 0.21
- B. 0.31
- C. 0.64
- D. 3.82

End of Section I

Turn over

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) The polynomial $P(x) = 4x^5 + 9x + 2$ is divided by $(x + 3)$. Find the remainder. **1**

(b) Solve the inequality $\frac{2}{2x + 3} \leq 1$. **3**

(c) Find $\int \frac{1}{\sqrt{9 - 4x^2}} dx$. **2**

(d) The polynomial $P(x) = x^4 + ax^3 - 3x^2 + bx - 2$ has a triple zero at $x = -1$. **3**
Find the values of a and b .

Question 11 continues on page 9

Question 11 (continued)

- (e) Evaluate $\int_0^{\frac{\pi}{2}} \cos^2(2\theta) d\theta$. **2**
- (f) Ten people sit around a round table. How many arrangements are possible if three particular people want to sit together? **1**
- (g) Find the term independent of x in the expansion of $\left(2x + \frac{1}{x^2}\right)^6$. **3**

End of Question 11

Question 12 (16 marks) Use a SEPARATE writing booklet.

- (a) A household iron is cooling in a room of constant temperature of 23°C . The initial temperature of the iron is 90°C and it cools to 70°C after 10 minutes.
You may assume that the rate of cooling can be expressed by the differential equation

$$\frac{dT}{dt} = -k(T - 23) \text{ where } k \text{ is a positive constant.}$$

- (i) Show, by differentiation, that the temperature T can be expressed in the form **1**

$$T = 23 + Ae^{-kt} \text{ where } A \text{ is a constant.}$$

- (ii) Find the **exact** values of A and k . **2**

- (iii) How long will it take for the iron to cool down to 40°C ? Give your answer to the nearest minute. **2**

- (b) (i) Differentiate $\tan^{-1}\left(\frac{1}{x}\right)$, $x \neq 0$, **and** hence show that **2**

$$\frac{d}{dx}\left(\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)\right) = 0 \text{ .}$$

- (ii) Hence explain why $\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}$ for $x > 0$. **2**

Question 12 (continued)

- (c) Functions f , g and h are defined such that

$$f(x) = \frac{1}{x-1} \quad , \quad g(x) = x^2 \quad , \quad h(x) = \sqrt{x} \quad .$$

- (i) State the equation of $f(h(x))$. **1**
- (ii) Determine the domain **and** range for $f(h(x))$. **4**
- (iii) Is it true that $f(h(g(x))) = f(x)$? Justify your answer. **2**

End of Question 12

Question 13 (14 marks) Use a SEPARATE writing booklet.

- (a) Use mathematical induction to prove that for all integers $n \geq 1$, **3**

$$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right)\cdots\left(1 - \frac{1}{(n+1)^2}\right) = \frac{n+2}{2n+2} \quad .$$

- (b) Find the **exact** volume of the solid of revolution formed when the graph of **2**

$x = \sqrt{\frac{1+2y}{1+y^2}}$ is rotated 360° about the y -axis for $y \in [0, 1]$.

- (c) Four runners have a race. All the runners complete the race. **3**

Determine the total number of ways they could finish the race, including the cases where two or more runners finish at exactly the same time.

Question 13 continues on page 13

Question 13 (continued)

(d)

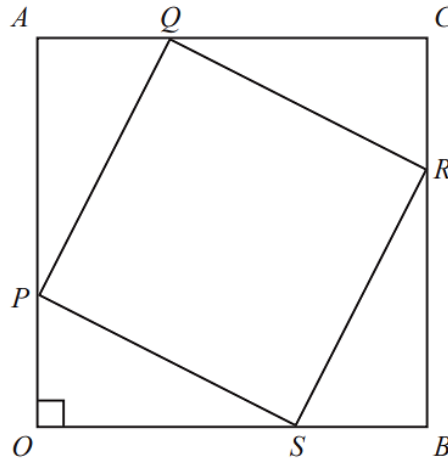


diagram not to scale

The diagram above shows the square $OACB$ where point O is the origin. Let the position vector for points A , B be defined as $\mathbf{a}$, $\mathbf{b}$ respectively, i.e. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

Let points P , Q , R and S be defined so that $\overrightarrow{OP} = \overrightarrow{RC} = k\mathbf{a}$ and $\overrightarrow{AQ} = \overrightarrow{SB} = k\mathbf{b}$ where $0 \leq k \leq 1$. This means that points P , Q , R and S are positioned along their respective sides in equal proportion.

(i) Explain why $\mathbf{a} \cdot \mathbf{b} = 0$. 1

(ii) Using vectors, prove that $\angle PQR = 90^\circ$. 2

Now suppose that in square $OACB$, it is known that $OA = 10$ cm and that point P is moving away from the origin at a constant speed of 0.2 cm per second.

This means that point P , Q , R and S are moving at the same speed along their respective sides.

Let x = the distance OP .

(iii) Determine the rate, in cm^2 per second, at which the area of square $PQRS$ is changing when $x = 3$ cm. 3

End of Question 13

Turn over

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) Solve $\sin x - 3 \cos x = 3$ for $0^\circ \leq x \leq 360^\circ$. **3**

- (b) An object is fired from a point O with initial speed of $V \text{ ms}^{-1}$ at an angle of elevation θ to the horizontal plane.

The position vector $\mathbf{s}(t)$ of at any time, t seconds, for $t \geq 0$, is

$$\mathbf{s}(t) = (Vt \cos \theta) \mathbf{i} + \left(Vt \sin \theta - \frac{1}{2}gt^2 \right) \mathbf{j},$$

where $g \text{ ms}^{-2}$ is the acceleration due to gravity, $\mathbf{i}$ is a unit vector in the horizontal direction of motion and $\mathbf{j}$ is a unit vector vertically up.

The object falls to a point P below the level of O such that $PM = OM$.

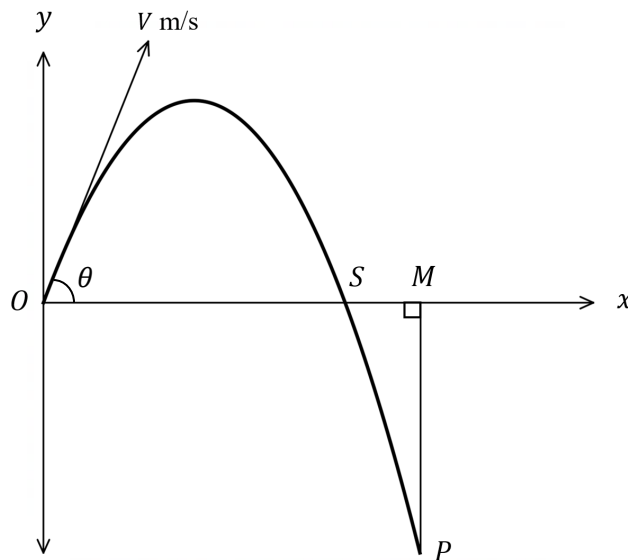


diagram not to scale

(i) Show that the distance OM is $\frac{V^2}{g} (\sin 2\theta + \cos 2\theta + 1)$ metres. **3**

(ii) Given that $\frac{OM}{OS} = \frac{4}{3}$, prove that $\sin 2\theta - 3 \cos 2\theta = 3$. **3**

(iii) Find the value of θ . **2**

Question 14 continues on page 15

Question 14 (continued)

- (c) The exact value of ${}^{100}\text{C}_3 + {}^{100}\text{C}_5 + \dots + {}^{100}\text{C}_{99}$ can be written in the form $a^b + c$ where $a, b, c \in \mathbb{Z}$. **4**

Find the values of a, b and c .

End of Question 14

End of Task

2022 HSC 12MAX Task 4 MC Solutions

$$1. \quad |z|^2 = |a|^2 + |b|^2 - 2|a||b|\cos 120^\circ \\ = |a|^2 + |b|^2 + |a||b|$$

$\therefore D$

$$2. \quad y = x^3 - 1$$

$$\text{inverse function : } x = y^3 - 1$$

$$\frac{dx}{dy} = 3y^2$$

$$\frac{dy}{dx} = \frac{1}{3y^2}$$

$$\text{at } (7, 2), \quad \frac{dy}{dx} = \frac{1}{3 \times 2^2} = \frac{1}{12}$$

$\therefore B$

$$3. \quad \text{domain } [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \\ \cos^{-1}(x) \rightarrow \cos^{-1}\left(\frac{3x}{\pi}\right)$$

$\therefore A$

$$4. \quad \cos \theta = \frac{a \cdot b}{|a||b|} \qquad \cos \varphi = \frac{b \cdot z}{|b||z|} \\ = \frac{x-1}{\sqrt{x^2+1} \times \sqrt{1+1}} \qquad = \frac{1-x}{\sqrt{1+1} \times \sqrt{1+x^2}} \\ = \frac{x-1}{\sqrt{2} \sqrt{x^2+1}} \qquad = \frac{1-x}{\sqrt{2} \sqrt{x^2+1}}$$

$$\cos \theta \cos \varphi = - \frac{(x-1)^2}{2(x^2+1)}$$

$\therefore B$

5. red blue green

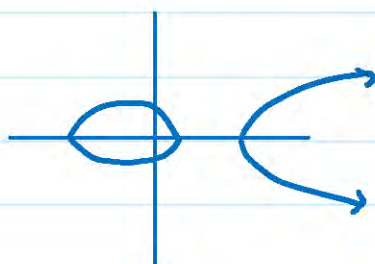
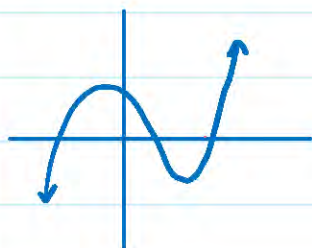
$$14 + 14 + 14 = 42$$

$$\text{at least 15 : } 42 + 1 = 43$$

$\therefore A$

6. $y = f(x)$

$y^2 = f(x)$



$\therefore A$

7. Method 1

$$\begin{array}{r} x^2 + x - 4 \\ x^2 + x - 2 \overline{) x^4 + 2x^3 - 5x^2 + x + 3} \\ \underline{x^4 + x^3 - 2x^2} \\ x^3 - 3x^2 + x \\ \underline{x^3 + x^2 - 2x} \\ -4x^2 + 3x + 3 \\ \underline{-4x^2 - 4x + 8} \\ 7x - 5 \end{array}$$

Method 2

$$P(x) = x^4 + 2x^3 - 5x^2 + x + 3$$

$$P(x) = (x^2 + x - 2)Q(x) + (Ax + B)$$

$$P(x) = (x+2)(x-1)Q(x) + (Ax + B)$$

$$P(1) = A + B$$

$$A + B = 1 + 2 - 5 + 1 + 3$$

$$\begin{array}{r} -4x + 5x + 7 \\ -4x^2 - 4x + 8 \\ \hline 7x - 5 \end{array}$$

$$P(1) = A + B$$

$$\begin{aligned} A + B &= 1 + 1 - 5 + 1 + 3 \\ &= 2 \end{aligned}$$

$$Ax + B = 7x - 5 \Rightarrow A + B = 2$$

$\therefore D$

8. $y = \sqrt{9 - (x-1)^2}$ semicircle with centre $(1, 0)$ and radius 3
 $\hookrightarrow$ domain: $[-2, 4]$

$$\begin{aligned} V &= \int_{-2}^4 9 - (x-1)^2 dx \quad \text{or} \quad V = \frac{4}{3} \pi r^3 \\ &= \frac{4}{3} \pi \times 3^3 \\ &= 4\pi \times 3^2 \end{aligned}$$

$\therefore A$

9. $x = \sec t + 1$ $\sec^2 t = (x-1)^2$
 $y = \tan t$ $\tan^2 t = y^2$

$$\begin{aligned} 1 + \tan^2 t &= \sec^2 t & \text{since } t &\in [0, \frac{\pi}{2}) \\ 1 + y^2 &= (x-1)^2 & x &\in [2, \infty) \\ y^2 &= x^2 - 2x & y &\in [0, \infty) \end{aligned}$$

$\therefore A$

10. $r = h \tan 60^\circ$ $\frac{dv}{dh} = 9\pi h^2$
 $= \sqrt{3} h$

$$= \sqrt{3} h$$

$$V = \pi r^2 h$$

$$= \pi (\sqrt{3} h)^2 h$$

$$= 3\pi h^3$$

$$\frac{dh}{dt} = 7\pi h$$

$$\frac{dV}{dt} = 1.5 \text{ m}^3/\text{min}$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$= \frac{1}{9\pi \times 0.5^2} \times 1.5$$

$$= 0.212206 \dots$$

$\therefore A$

2022 HSC 12MAX Task 4 Question 11 Solutions

Question 11

$$(a) \quad P(-3) = 4 \times (-3)^5 + 9 \times (-3) + 2 \\ = -997 \quad (1)$$

$$(b) \quad \frac{2}{2x+3} \leq 1 \quad x \neq -\frac{3}{2}$$

$$x(2x+3)^2 \quad x(2x+3)^2$$

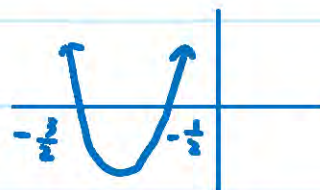
$$2(2x+3) \leq (2x+3)^2 \quad (1)$$

$$(2x+3)^2 - 2(2x+3) \geq 0$$

$$(2x+3)(2x+3-2) \geq 0$$

$$(2x+3)(2x+1) \geq 0$$

$$\therefore x < -\frac{3}{2}, \quad x \geq -\frac{1}{2} \quad (1)$$



$$(c) \quad \int \frac{1}{\sqrt{9-4x^2}} dx$$

$$= \frac{1}{2} \int \frac{2}{\sqrt{9-4x^2}} dx$$

$$= \frac{1}{2} \sin^{-1}\left(\frac{2x}{3}\right) + C \quad (1)$$

$$(d) \quad 2 \text{ is the other root since } (-1)^3 + 2 = -2 \\ a = \dots \quad (1)$$

(a) 2 is the other root since $(-1)^4 + a(-1) - 2 = -a$

$$-\frac{a}{1} = -1 + -1 + -1 + 2 \quad (1)$$

$$-a = -1$$

$$a = 1$$

$$P(x) = x^4 + x^3 - 3x^2 + bx - 2$$

$$P(-1) = 1 - 1 - 3 - b - 2 = 0$$

$$b = -5$$

$$\therefore a = 1 \quad b = -5 \quad (1)$$

$$(e) \int_0^{\frac{\pi}{2}} \cos^2(2\theta) d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 4\theta) d\theta \quad (1)$$

$$= \frac{1}{2} \left[\theta + \frac{1}{4} \sin 4\theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left(\frac{\pi}{2} + \frac{1}{4} \sin 2\pi \right)$$

$$= \frac{\pi}{4} \quad (1)$$

$$(f) 7! \times 3! = 30240 \quad (1)$$

$$(g) {}^6C_r (2x)^{6-r} (x^{-2})^r = {}^6C_r \times 2^{6-r} \times x^{6-r} \times x^{-2r} \quad (1)$$

$$= {}^6C_r \times 2^{6-r} \times x^{6-3r}$$

$$= {}^6C_r \times 2^{6-r} \times x^{6-r}$$

$$6 - 3r = 0$$

$$r = 2 \quad (1)$$

$$\therefore \text{term} = {}^6C_2 \times 2^{6-2}$$

$$= 240$$

(1)

2022 HSC 12MAX Task 4 Question 12 Solutions

Question 12

(a) (i) $T = 23 + Ae^{-kt}$

$$\frac{dT}{dt} = -k(Ae^{-kt}) \quad (1)$$

$$= -k(T - 23)$$

(ii) $t = 0, T = 90 \Rightarrow \begin{cases} 90 = 23 + Ae^{-k \times 0} \\ t = 10, T = 70 \end{cases} \quad \begin{cases} 70 = 23 + Ae^{-k \times 10} \end{cases}$

$$23 + A = 90$$

$$A = 67 \quad (1)$$

$$67e^{-10k} = 47$$

$$e^{-10k} = \frac{47}{67}$$

$$10k = \ln \frac{67}{47}$$

$$k = \frac{1}{10} \ln \frac{67}{47} \quad (1)$$

(iii) $T = 23 + 67e^{-\frac{1}{10} \ln \frac{67}{47} t}$

$$40 = 23 + 67e^{\frac{1}{10} \ln \frac{47}{67} t} \quad (1)$$

$$e^{\frac{1}{10} \ln \frac{47}{67} t} = \frac{17}{67}$$

$$\frac{1}{10} \ln \frac{47}{67} t = \ln \frac{17}{67}$$

$$t = 10 \ln \frac{17}{67} \div \ln \frac{47}{67}$$

$$t \approx 39 \text{ mins} \quad (1)$$

$$(b) (i) \text{ LHS} = \frac{1}{1+x^2} + \frac{-x^{-2}}{1+(x^{-1})^2} \quad (1)$$

$$= \frac{1}{1+x^2} + \frac{-x^{-2}}{1+x^{-2}}$$

$$= \frac{1}{1+x^2} + \frac{-1}{x^2+1} \quad (1)$$

$$= 0$$

$$= \text{RHS}$$

$$(ii) \text{ let } f(x) = \tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right)$$

$$f'(x) = 0 \quad (\text{from part (a)})$$

$$\therefore f(x) \text{ is a constant} \quad (1)$$

$$x = 1, \quad f(1) = \tan^{-1} 1 + \tan^{-1} 1$$

$$= \frac{\pi}{4} + \frac{\pi}{4} \quad (1)$$

$$= \frac{\pi}{2}$$

$$\therefore f(x) = \frac{\pi}{2} \text{ for } x > 0$$

$$(c) (i) f(h(x)) = f(\sqrt{x})$$

$$= \frac{1}{\sqrt{x}-1} \quad (1)$$

$$(ii) \text{ domain: } x \geq 0 \text{ and } x \neq 1$$

$$\therefore x \in [0, 1) \cup (1, \infty) \quad (1) \quad (1)$$

$$\text{range: } \sqrt{x} \geq 0$$

$$\sqrt{x} - 1 > -1$$

$$\text{range : } \sqrt{x} \geq 0$$

$$\sqrt{x} - 1 \geq -1$$

$$\downarrow \qquad \downarrow$$

$$-1 \leq \sqrt{x} - 1 < 0 \qquad \sqrt{x} - 1 > 0$$

$$\frac{1}{\sqrt{x} - 1} \leq -1 \qquad \frac{1}{\sqrt{x} - 1} > 0$$

$$\therefore f(h(x)) \in (-\infty, -1] \cup (0, \infty)$$

$$\begin{aligned} \text{(iii)} \quad h(g(x)) &= \sqrt{x^2} \\ &= |x| \end{aligned}$$

$$f(h(g(x))) = f(|x|) \neq f(x) \quad \text{since } f(x) \text{ is not even}$$

$\therefore$ the statement is false

2022 HSC 12MAX Task 4 Question 13 Solutions

Question 13

(a) ① Prove true for $n = 1$.

$$\begin{aligned} \text{LHS} &= 1 - \frac{1}{(1+1)^2} & \text{RHS} &= \frac{1+2}{2 \times 1 + 2} \\ &= \frac{3}{4} & &= \frac{3}{4} \end{aligned}$$

$\text{LHS} = \text{RHS} \Rightarrow$ the statement is true for $n = 1$

② Assume true for $n = k$, $k \in \mathbb{Z}^+$

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{(k+1)^2}\right) = \frac{k+2}{2k+2}$$

③ Prove true for $n = k+1$, that is

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{(k+1)^2}\right) \left(1 - \frac{1}{(k+2)^2}\right) = \frac{k+3}{2k+4}$$

$$\text{LHS} = \frac{k+2}{2k+2} \times \left(1 - \frac{1}{(k+2)^2}\right) \quad \text{from step 2}$$

$$= \frac{k+2}{2k+2} \times \frac{k^2 + 4k + 3}{(k+2)^2}$$

$$= \frac{k+2}{2(k+1)} \times \frac{(k+1)(k+3)}{(k+2)^2}$$

$$= \frac{k+3}{2(k+2)}$$

$$= \frac{k+3}{2k+4}$$

$\therefore$ if the statement is true for $n = k$,

$= \text{RHS}$ then it is true for $n = k+1$

④ The statement is true for $n = 1$. Whenever (or if) the statement is true for $n = k$ it is also true

Ⓔ The statement is true for $n=1$. Whenever (or if) the statement is true for $n=k$, it is also true for $n=k+1$. Hence, by principle of mathematical induction, the statement is true for positive integers $n \geq 1$.

$$\begin{aligned}
 \text{(b)} \quad V &= \pi \int_0^1 \frac{1+2y}{1+y^2} dy \\
 &= \pi \int_0^1 \frac{1}{1+y^2} + \frac{2y}{1+y^2} dy \\
 &= \pi [\tan^{-1}y + \ln|1+y^2|]_0^1 \\
 &= \pi (\tan^{-1}1 + \ln 2) \\
 &= \frac{\pi^2}{4} + \pi \ln 2 \quad \text{units}^3
 \end{aligned}$$

(c) four different : $4!$
 three different : ${}^4C_2 \times 3!$
 two different : 4C_2
 one different : ${}^4C_3 \times 2!$
 all same : 1

$$\begin{aligned}
 \text{total} &= 4! + {}^4C_2 \times 3! + {}^4C_2 + {}^4C_3 \times 2! + 1 \\
 &= 75
 \end{aligned}$$

(d) (i) $\vec{OA} \perp \vec{OB}$ therefore $\underline{a} \cdot \underline{b} = 0$

(ii) $\vec{PQ} = \vec{PO} + \vec{OA} + \vec{AQ}$

$$= -k\underline{a} + \underline{a} + k\underline{b}$$

$$= (1-k)\underline{a} + k\underline{b}$$

$$\vec{QR} = \vec{QA} + \vec{AC} + \vec{CR}$$

$$= -k\underline{b} + \underline{b} + -k\underline{a}$$

$$= (1-k)\underline{b} - k\underline{a}$$

$$\vec{PQ} \cdot \vec{QR} = ((1-k)\underline{a} + k\underline{b}) \cdot ((1-k)\underline{b} - k\underline{a})$$

$$= (1-k)^2 \underline{a} \cdot \underline{b} - k(1-k)|\underline{a}|^2 + k(1-k)|\underline{b}|^2 - k^2 \underline{a} \cdot \underline{b}$$

since ABCD is a square,

$$\underline{a} \cdot \underline{b} = 0 \quad \text{and} \quad |\underline{a}| = |\underline{b}|$$

$$\vec{PQ} \cdot \vec{QR} = -k(1-k)|\underline{a}|^2 + k(1-k)|\underline{b}|^2$$

$$= -k(1-k)|\underline{a}|^2 + k(1-k)|\underline{a}|^2$$

$$= 0$$

$$\therefore \angle PQR = 90^\circ$$

(iii) $x = OP$

$$A_{PQRS} = 10^2 - 4 \times \frac{1}{2} \times x(10-x)$$

$$\frac{dx}{dt} = 0.2 \text{ cm/s}$$

$$= 100 - 2x(10-x)$$

$$= 2x^2 - 20x + 100$$

$$\frac{dA}{dx} = 4x - 20$$

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

$$= (4x - 20) \times 0.2$$

$$= (4 \times 3 - 20) \times 0.2$$

$$= -1.6 \text{ cm}^2 / \text{s}$$

Question 14

$$(a) \quad \sin x - 3 \cos x = R \sin(x - \alpha)$$

$$\begin{cases} R \cos \alpha = 3 \\ R \sin \alpha = 1 \end{cases} \Rightarrow \begin{cases} R = \sqrt{10} \\ \alpha = \tan^{-1}(1/3) \end{cases}$$

$$\sqrt{10} \sin(x - \tan^{-1}(1/3)) = 3$$

$$\sin(x - \tan^{-1}(1/3)) = \frac{3\sqrt{10}}{10}$$

$$x - \tan^{-1}(1/3) = 71.56505 \dots^\circ \text{ or } 108.4349 \dots^\circ$$

$$x = 143^\circ \text{ or } 180^\circ$$

$$(b) (i) \text{ at } P, \quad Vt \cos \theta = \frac{1}{2}gt^2 - Vt \sin \theta$$

$$2Vt \cos \theta = gt^2 - 2Vt \sin \theta$$

$$-gt^2 + 2Vt(\cos \theta + \sin \theta) = 0$$

$$t(-gt + 2V(\cos \theta + \sin \theta)) = 0$$

$$t = 0, \quad -gt + 2V(\cos \theta + \sin \theta) = 0$$

$$gt = 2V(\cos \theta + \sin \theta)$$

$$t = \frac{2V(\cos \theta + \sin \theta)}{g}$$

$$t = \frac{2V(\cos \theta + \sin \theta)}{g}, \quad x = V \cos \theta \times \frac{2V(\cos \theta + \sin \theta)}{g}$$

$$= \frac{2V^2(\cos^2 \theta + \cos \theta \sin \theta)}{g}$$

$$= \frac{V^2}{g} (2\cos^2 \theta + 2\sin \theta \cos \theta)$$

$$= \frac{V^2}{g} (\cos 2\theta + 1 + \sin 2\theta)$$

$$= \frac{V^2}{g} (\sin 2\theta + \cos 2\theta + 1)$$

$$(ii) \text{ OS: } Vt \sin \theta - \frac{1}{2}gt^2 = 0$$

$$2Vt \sin \theta - gt^2 = 0$$

$$t(2V \sin \theta - gt) = 0$$

$$t = \frac{2V \sin \theta}{g}$$

$$t(2V \sin \theta - gt) = 0$$

$$t = 0, \quad t = \frac{2V \sin \theta}{g}$$

$$x = V \cos \theta \times \frac{2V \sin \theta}{g}$$

$$= \frac{V^2}{g} \sin 2\theta$$

$$\frac{OM}{OS} = \frac{4}{3} \Rightarrow \frac{\frac{V^2}{g} (\sin 2\theta + \cos 2\theta + 1)}{\frac{V^2}{g} \sin 2\theta} = \frac{4}{3}$$

$$\frac{\sin 2\theta + \cos 2\theta + 1}{\sin 2\theta} = \frac{4}{3}$$

$$3 \sin 2\theta + 3 \cos 2\theta + 3 = 4 \sin 2\theta$$

$$\sin 2\theta - 3 \cos 2\theta = 3$$

$$(ii) \quad 2\theta = 143^\circ, \quad 180^\circ$$

$$\theta = 71.5^\circ, \quad 90^\circ$$

Since θ is angle of elevation,

$$\theta = 71.5^\circ$$

$$(c) \quad (1+x)^{100} - (1-x)^{100}$$

$$= {}^{100}C_0 x^0 + {}^{100}C_1 x^1 + {}^{100}C_2 x^2 + \dots + {}^{100}C_{99} x^{99} + {}^{100}C_{100} x^{100}$$

$$- ({}^{100}C_0 x^0 - {}^{100}C_1 x^1 + {}^{100}C_2 x^2 - \dots - {}^{100}C_{99} x^{99} + {}^{100}C_{100} x^{100})$$

$$= 2 \times ({}^{100}C_1 x^1 + {}^{100}C_3 x^3 + \dots + {}^{100}C_{99} x^{99})$$

$$\text{Sub } x = 1,$$

$$\text{LHS} = 2^{100} - 0^{100}$$

$$= 2^{100}$$

$$\text{RHS} = 2 \times ({}^{100}C_1 + {}^{100}C_3 + \dots + {}^{100}C_{99})$$

$$= 2 \times (100 + {}^{100}C_3 + \dots + {}^{100}C_{99})$$

$$2^{100} = 2 \times (100 + {}^{100}C_3 + \dots + {}^{100}C_{99})$$

$$2^{99} = 100 + {}^{100}C_3 + \dots + {}^{100}C_{99}$$

$${}^{100}C_3 + \dots + {}^{100}C_{99} = 2^{99} - 100$$

$$2^{100} = 100 + {}^{100}C_3 + \dots + {}^{100}C_{99}$$

$${}^{100}C_3 + \dots + {}^{100}C_{99} = 2^{99} - 100$$